September 1, 2026

Airbus Operations GmbH
Hamburg

Dear Hiring Team,

I am writing to apply for the Fatigue & Damage Tolerance Engineer position in Hamburg. I am a structural analysis and simulation engineer with over seven years in aerospace structural integrity, spanning fatigue and damage tolerance of composite, hybrid, and metallic structures, and I would bring that depth to a role framed as an entry point into F&DT.

Analyzing and validating structures against defined F&DT methods is the core of my day-to-day work. During my PhD at DLR and TU Braunschweig I derived a certification-aligned failure criterion, a leakage envelope, that maps combined strain and load states to failure onset and defines an allowable-damage-limit philosophy consistent with regulatory boundary conditions. As a Research Intern in Metallic Fatigue and Damage Tolerance at DLR, I coupled defect measurements with numerical simulation to establish a defect-tolerance qualification methodology, translating fracture-mechanics and fatigue criteria into allowable-defect acceptance limits, the stress-analysis basis for accepting or rejecting a defective structure. I also validate theoretical damage models against measured data through Monte Carlo uncertainty quantification, and I work with progressive damage analysis of composites (CDM, PDA, CZM) alongside metallic fatigue assessment.

Beyond the analysis itself, I regularly work across design, materials, and stress interfaces: I provide sign-off-style technical review of externally and student-produced structural analysis work, define technical scope for that work, and communicate findings to design and project teams. I work daily with the FE solvers (Nastran/Patran, ABAQUS) and analytical stress methods (Niu, Bruhn) that sit alongside F&DT sizing, build Python-based automation for analysis and reporting, and am genuinely interested in applying AI to improve engineering workflows.

To be transparent about fit: my acceptance-limit work has been for manufacturing-induced defects rather than in-service damage, so I have not designed or validated physical repair schemes, and I have not used Hyperworks; my FE tool experience is Nastran/Patran and ABAQUS. My German is currently at B1 level. I would rather name these directly than have them surface later.

I am motivated by the chance to bring rigorous, certification-oriented F&DT thinking to Airbus, and I would welcome the opportunity to discuss how my background in damage-tolerance philosophy, model verification, and technical review can contribute to your team.

Thank you for your consideration. I look forward to discussing how I can contribute.

Sincerely,

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